03/23 Monday ~~Friday~~.

$$\frac{y_0 - 4000}{150} = 3.10.$$

$$y_0 = 3.10 \times 150 + 4000 = \overset{\shortparallel}{V}.$$

<u>Theorem</u>: Suppose $X_1, \dots X_n$ are independent R.Vs with mean μ, and variance σ^2; i.e.,

$$E(X_i) = \mu \qquad Var(X_i) = \sigma^2. \qquad \text{for } i = 1, \dots n.$$

Let $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i = \frac{1}{n}X_1 + \frac{1}{n}X_2 + \dots + \frac{1}{n}X_n.$

then. 1) $\mu_{\bar{X}} = E(\bar{X}) = \mu.$

2) $\sigma_{\bar{X}}^2 = Var(\bar{X}) = \frac{\sigma^2}{n}.$

<u>Proof</u>: $E(\bar{X}) = E(\frac{1}{n}X_1 + \dots + \frac{1}{n}X_n)$

$$= E(\tfrac{1}{n}X_1) + E(\tfrac{1}{n}X_2) + \dots + E(\tfrac{1}{n}X_n)$$

$$= \frac{1}{n}\left[E(X_1) + E(X_2) + \dots E(X_n)\right]$$

$$= \frac{1}{n}\left[\underbrace{\mu + \mu + \dots + \mu}_{n.}\right] = \mu.$$

$$Var(\bar{X}) = Var(\tfrac{1}{n}X_1 + \dots + \tfrac{1}{n}X_n).$$

$$\overset{ind.}{=} (\tfrac{1}{n})^2 Var(X_1) + \dots (\tfrac{1}{n})^2 Var(X_n)$$

$$= \frac{1}{n^2}\left[Var(X_1) + \dots + Var(X_n)\right]$$

$$= \frac{1}{n^2}\left[\underbrace{\sigma^2 + \sigma^2 + \dots + \sigma^2}_{n.}\right]$$

$$= \frac{1}{n^{\cancel{2}}} \cdot \cancel{n}\sigma^2 = \frac{\sigma^2}{n}$$

Theorem: Suppose $X_1 \ldots X_n$ are independent R.Vs where $X_i \overset{iid}{\sim} N(\mu, \sigma^2)$

then $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i \sim N(\mu_{\bar{X}} = \mu, \sigma^2_{\bar{X}} = \frac{\sigma^2}{n})$

Example: Suppose 25 people are R.S. (independent)

$X_i \overset{iid}{\sim} N(\mu = 160, \sigma^2 = 30^2)$ $\quad i = 1, \ldots 25.$

a) distribution of $\bar{X}$.

$\bar{X} \sim N(\mu_{\bar{X}} = 160, \sigma^2_{\bar{X}} = \frac{30^2}{25} = 6^2)$

b) probability $\bar{X} > 166$

$$\underbrace{P(\bar{X} > 166)} = P\left(\boxed{\frac{\bar{X} - 160}{6}} > \frac{166 - 160}{6}\right)$$

$$= P(Z > 1)$$

$$= 1 - P(Z \leq 1) = 1 - 0.8413 = 0.1587$$

c) probality 25 people together larger than 4500 pounds.

$$P\left(\sum_{i=1}^{n} X_i > 4500\right) = P\left(\frac{1}{n}\sum_{i=1}^{n} X_i > \frac{1}{n} 4500\right)$$

$$= \underbrace{P\left(\bar{X} > \frac{4500}{25}\right)}$$

Delta method (Propagation of error)

Recall: Taylor series.

Suppose $u(x)$ is infinitely differentiable at a constant "a"

then $u(x) = u(a) + \boxed{\left.\frac{du}{dx}\right|_{x=a}} \frac{(x-a)^1}{1!} + \left.\frac{d^2u}{dx^2}\right|_{x=a} \frac{(x-a)^2}{2!} + \cdots$

$\approx u(a) + u^{(1)}(a)(x-a) + \frac{u^{(2)}(a)}{2}(x-a)^2$

$f(x)$

$f'(x)$ $f^{(1)}(x)$

$f''(x)$ $f^{(2)}(x)$

Note: We can approximate function of R.Vs in a similar way.

Theorem: Suppose $X \sim \cdot(\mu_x, \sigma_x^2)$.

Taylor approximation about the mean μ_x is.

$$u(X) \approx u(\mu_x) + u^{(1)}(\mu_x)(X-\mu_x) + \frac{u^{(2)}(\mu_x)}{2}(X-\mu_x)^2.$$

Now. $\boxed{\mu_u} = E[u(X)] \approx E\left[u(\mu_x) + u^{(1)}(\mu_x)(X-\mu_x) + \frac{u^{(2)}(\mu_x)}{2}(X-\mu_x)^2\right]$

$= u(\mu_x) + u^{(1)}(\mu_x)\underbrace{E(X-\mu_x)}_{\substack{= EX - E\mu_x \\ = \mu_x - \mu_x = 0}} + \frac{u^{(2)}(\mu_x)}{2}\underbrace{E(X-\mu_x)^2}_{\sigma_x^2}$

$$= \boxed{u(\mu_x) + \frac{u^{(2)}(\mu_x)}{2}\sigma_x^2}$$